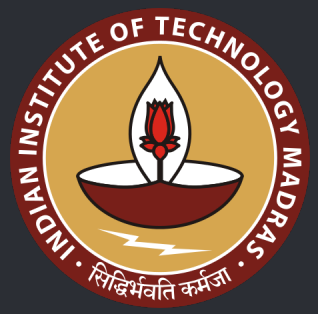


Dynamic Traffic Signal Control using Reinforcement Learning

Akash Ankush Kamble [CS17M018]

Guide: Dr. L. A. Prashanth

Section 1

1. Introduction
2. The Signal Control Problem
   - 2.1 MDP framework
   - 2.2 Traffic Signal Control as an MDP
3. Algorithms
   - 3.1 Monte Carlo Control
   - 3.2 Q-learning
4. Experimental Setup
5. Results
6. Conclusions and Future Work

The traffic problem

- It is getting worse!

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- Crossroad junctions are a bottleneck.

The traffic problem

- It is getting worse!
- Crossroad junctions are a bottleneck.
- Fixed-time controllers are inefficient.

Our Goal[1]

- Testing reinforcement learning (RL)-based method(s) for TSC

[1] A similar problem is studied in [2]

Our Goal[1]

- Testing reinforcement learning (RL)-based method(s) for TSC
- Traffic flow maximization

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Our Goal[1]

- Testing reinforcement learning (RL)-based method(s) for TSC
- Traffic flow maximization
- Waiting time (or delay) minimization

[1] A similar problem is studied in [2]

Our Goal[1]

- Testing reinforcement learning (RL)-based method(s) for TSC
- Traffic flow maximization
- Waiting time (or delay) minimization
- i.e., Finding the best policy for switching traffic lights.

[1] A similar problem is studied in [2]

Section 2

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The State, Action and Reward

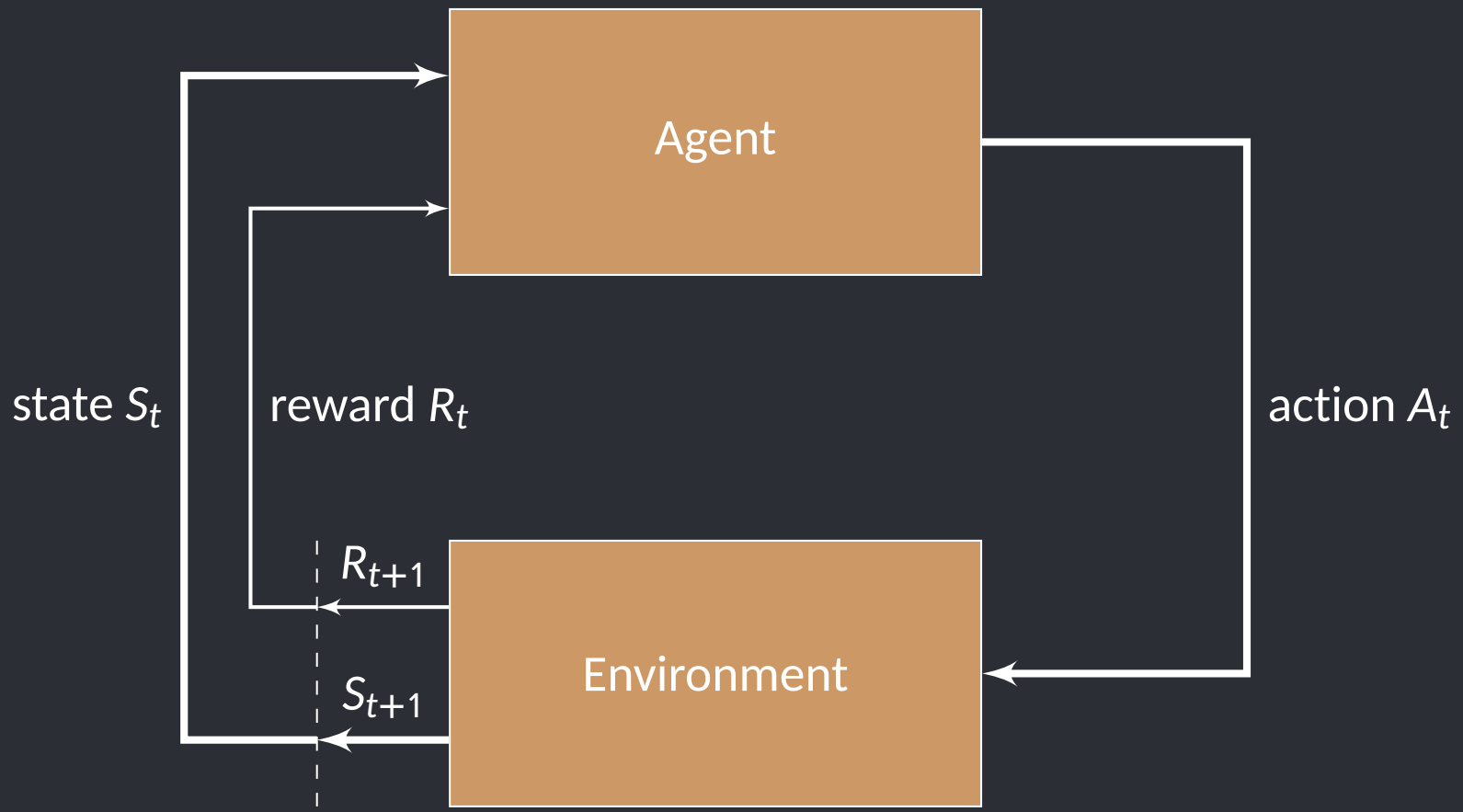


Figure: The agent-environment interrelationship in a MDP.[2]

[2] Figure redrawn from section 3.1 of [3]

State, Action and Reward for full-state-based algorithms [MCC, QLFS]

State

- An L dimensional vector of queue lengths
- Three categories of queue lengths:
  - Low
  - Medium
  - High
- State at time t is,

$$s_t = (q_1(t), q_2(t), \ldots, q_L(t))$$

Action

- The action set has duration values for all lanes of all phases
- Action vector can be represented as,

$$a = ((d_1^1, d_2^1, \ldots, d_K^1), (d_1^2, d_2^2, \ldots, d_K^2), \ldots, (d_1^N, d_2^N, \ldots, d_K^N))$$

where, d_j^i is the jth available duration choice for the ith traffic light phase.

Reward

- Sum of individual lane rewards
- Calculated as,

$$r(s_t, a_t) = r_1 + r_2 + \cdots + r_N$$

- Lane rewards are set as follows:

Lane occupancy percentage	Traffic category	Reward
0 - 29	Low	0
30 - 59	Medium	-1
60 - 100	High	-2

State, Action and Reward for feature-based algorithm [QLFA]

State

- A vector of size ($d = |F| * |A| + 1$) populated with feature values, per action

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- Extracted two features (1 for queue length, 1 for time)

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  - The queue lengths are not categorized, but summed up directly and normalized to one real number $\in [0, 1]$

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  - The time feature records elapsed time since last green/red switch and normalizes again to a real number $\in [0, 1]$

State

- A vector of size ($d = |F| * |A| + 1$) populated with feature values, per action
- Extracted two features (1 for queue length, 1 for time)
  - The queue lengths are not categorized, but summed up directly and normalized to one real number $\in [0, 1]$
  - The time feature records elapsed time since last green/red switch and normalizes again to a real number $\in [0, 1]$
- The rest of the terms in feature vector are zeros.

State

Given n actions and two extractable features per action, the features for state-action pair (s, a_i) can be encoded as:

$$\phi(s,a_1) = \begin{bmatrix} \psi_1(s,a_1) \\ \psi_2(s,a_1) \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \\ 1 \end{bmatrix}, \phi(s,a_2) = \begin{bmatrix} 0 \\ 0 \\ \psi_1(s,a_2) \\ \psi_2(s,a_2) \\ \vdots \\ 0 \\ 0 \\ 1 \end{bmatrix}, \cdots, \phi(s,a_n) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \vdots \\ \psi_1(s,a_n) \\ \psi_2(s,a_n) \\ 1 \end{bmatrix}$$

So, Q is approximated as $Q(s,a) \approx \theta^T \phi(s,a)$

Action

(No change in action vector)

- The action set has duration values for all lanes of all phases
- Action vector can be represented as,

$$a = ((d_1^1, d_2^1, \ldots, d_K^1), (d_1^2, d_2^2, \ldots, d_K^2), \ldots, (d_1^N, d_2^N, \ldots, d_K^N))$$

where, d_j^i is the jth available duration choice for the ith traffic light phase.

Reward

- No categories

Reward

- No categories
- Penalty based on sum of feature values

Reward

- No categories
- Penalty based on sum of feature values
- Calculated as,

$$r(s_t, a_t) = \left[\sum_{i=1}^{d} \phi_{s_t, a_t}(i) \right] * (-1)$$

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MCC [3]

Algorithm 1 : Monte Carlo Control (MCC)

Input: Algorithm parameter: $\epsilon > 0$, episode length: T
Initialize: π, $Q(s, a)$, $Returns(s, a)$
for all episodes **do**
    Produce an episode following π: $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T$
    Set $G \leftarrow 0$
    for all steps of the episode, $t = T-1, T-2, \ldots, 0$ **do**
        $G \leftarrow G + R_{t+1}$
        if the pair S_t, A_t appears in episode **then**
            Append G to $Returns(S_t, A_t)$
            $Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$
            $A^* \leftarrow \arg\max_a Q(S_t, a)$
            for all $a \in \mathscr{A}(S_t)$ **do**

$$\pi^*(a|S_t) \leftarrow \begin{cases} 1-\epsilon & \text{, if } a = A^* \\ \epsilon & \text{, if } a \neq A^* \end{cases}$$

[3] see Section 5.4 of [3] for a detailed description

Q-learning[4]

Algorithm 2 : Q-learning

Input: Algorithm parameters: $\alpha \in (0, 1]$, $\epsilon > 0$, episode length: T
Initialize: π, $Q(s, a)$
for all episodes **do**
    Initialize S
    for all steps of the episode **do**
        Choose A from S using ϵ-greedy policy derived from Q
        Take action A, observe R, S'
        $Q(S, A) \leftarrow Q(S, A) + \alpha[R + \gamma max_a Q(S', a) - Q(S, A)]$
        $S \leftarrow S'$

[4] see Section 6.5 of [3] for a detailed description

Q-learning

The modified update for QLFA:

$$\theta \leftarrow \theta + \alpha \left[R + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \sum_{i=1}^{|A|} \phi_i(s, a)$$

where,

- $\alpha \in [0, 1]$ is the learning rate or step size,
- $\gamma \in [0, 1]$ is the discount factor,
- $\max_{a'} Q(s', a')$ is the action a' in the state s' maximizing $Q(s', a)$,

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SUMO[5]

Traffic **C**ontrol **I**nterface (TraCI):

- Library of commands and methods
- Control over the simulation and the actions of objects within it
- TCP based client-server design for communication

[5] **S**imulation of **U**rban **MO**bility, see [1]

SUMO GUI

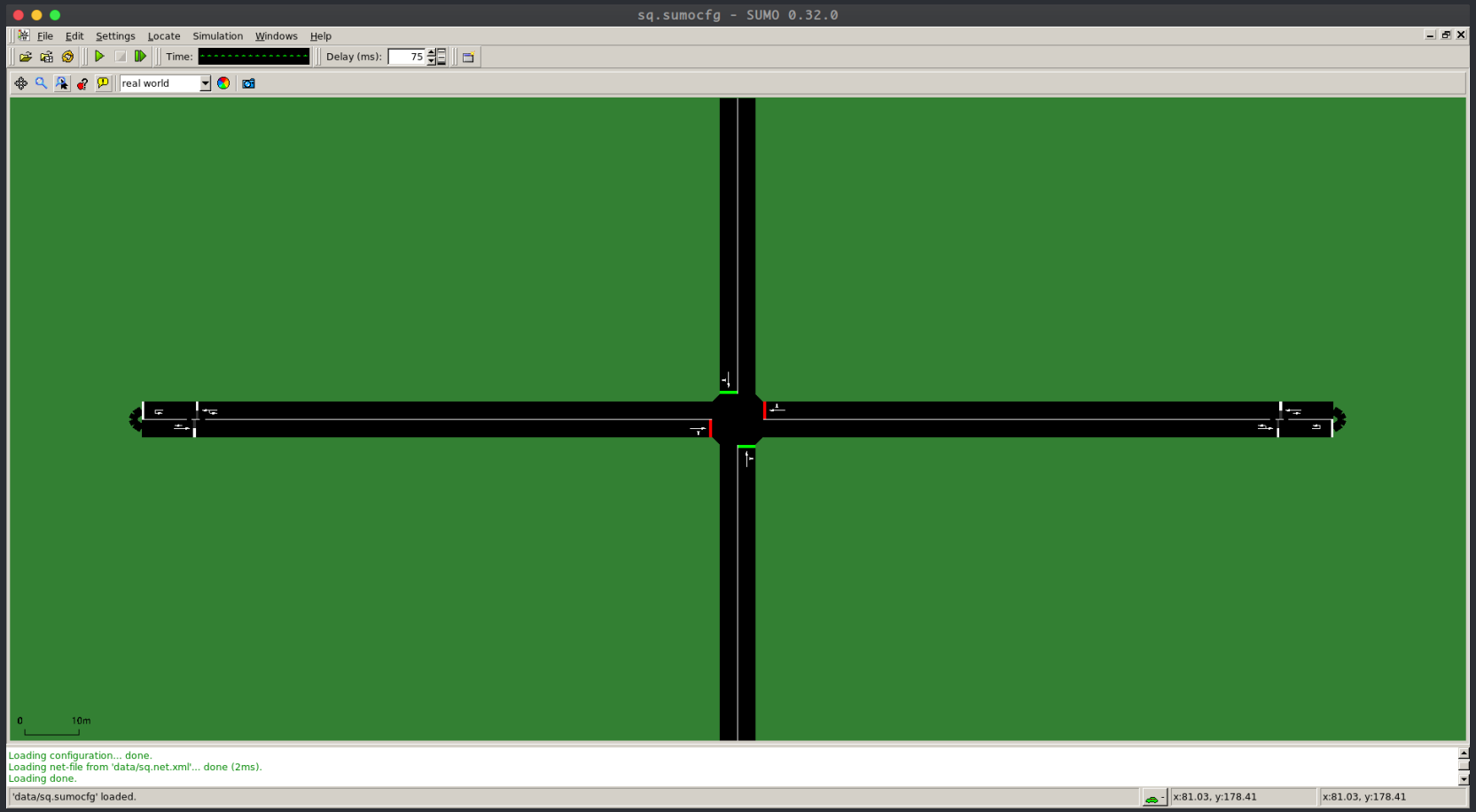


Figure: SUMO - Graphical User Interface

Old experimental setup

- Q-values: *Random numbers* $\in (0, 1]$
- Number of vehicles: 7942
- Episode length (T): 850 timesteps
- Exploration factor (ϵ): 0.2
- Discount factor (γ): 0.9
- Step size (α): 0.05

Current experimental setup

Scenario 1: a simple junction(two crossroads)

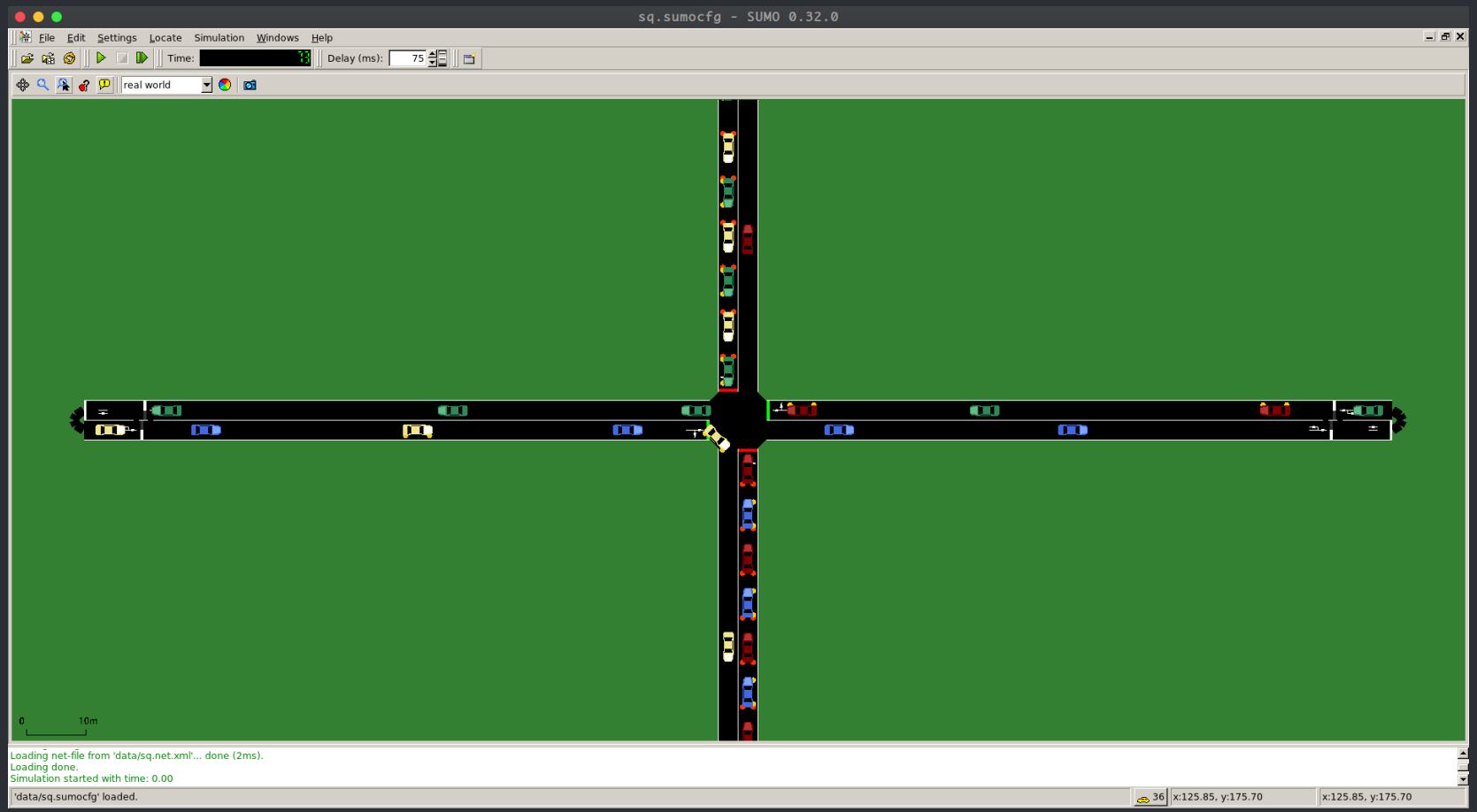


Figure: Single junction

Scenario 1: a simple junction(two crossroads)

Parameters for the algorithms

- Q-values: *Random numbers* $\in (0, 1]$
- Φ matrix, θ vector: *Random numbers* $\in (0, 1]$
- Number of vehicles: 7942
- Episode length (T): 75 timesteps
- Exploration factor (ϵ): 0.2
- Discount factor (γ): 0.9
- Step size (α): 0.05

Scenario 2: two-junction corridor

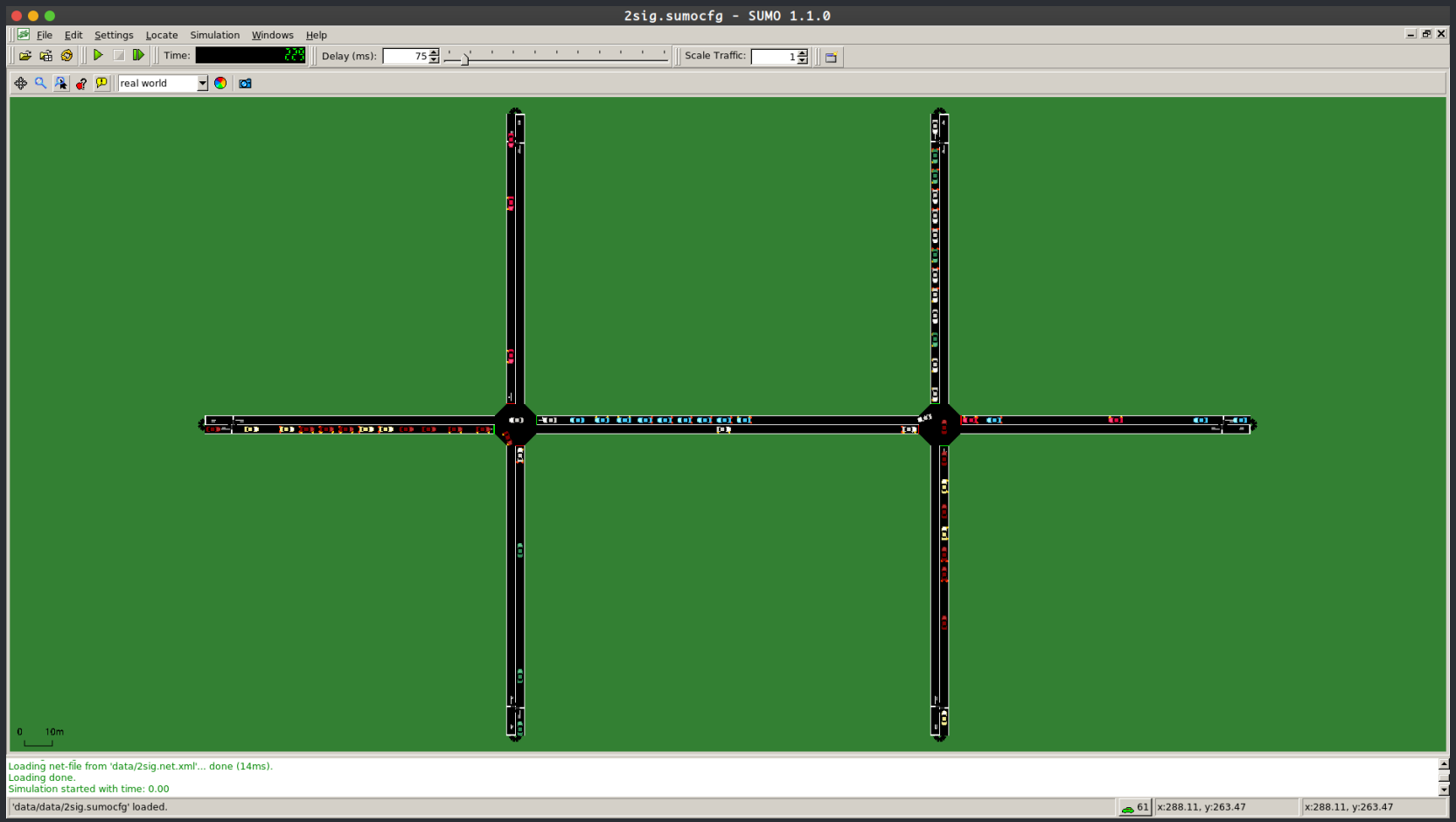


Figure: Two-junction corridor

Scenario 2: two-junction corridor

Parameters for the algorithms

- Q-values: *Random numbers* $\in (0, 1]$
- Φ matrix, θ vector: *Random numbers* $\in (0, 1]$
- Number of vehicles: 9951
- Episode length (T): 300 timesteps
- Exploration factor (ϵ): 0.2
- Discount factor (γ): 0.9
- Step size (α): 0.05

Scenario 3: a 2×2 grid

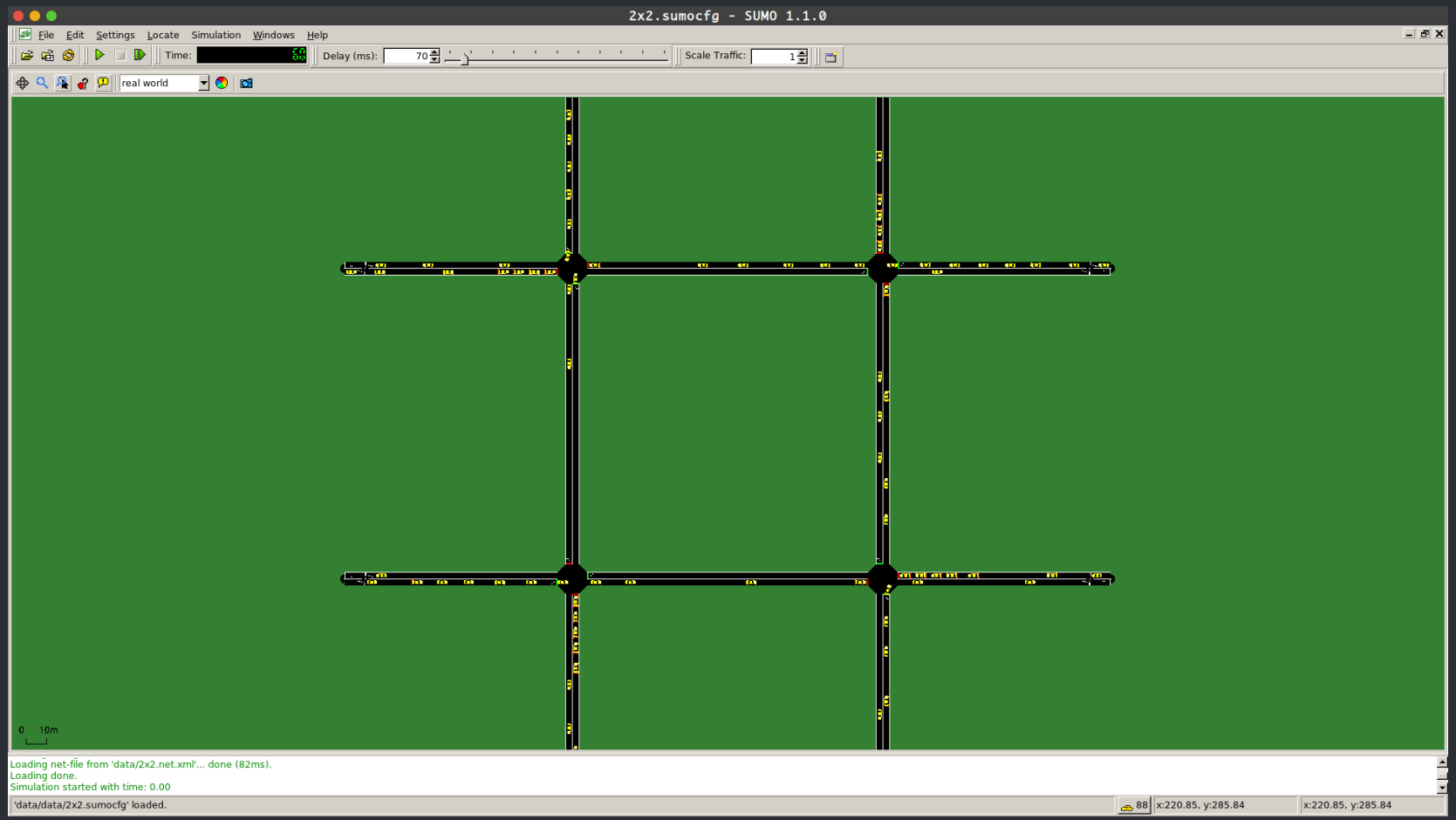


Figure: 2 × 2 grid

Scenario 3: a 2×2 grid

Parameters for the algorithms

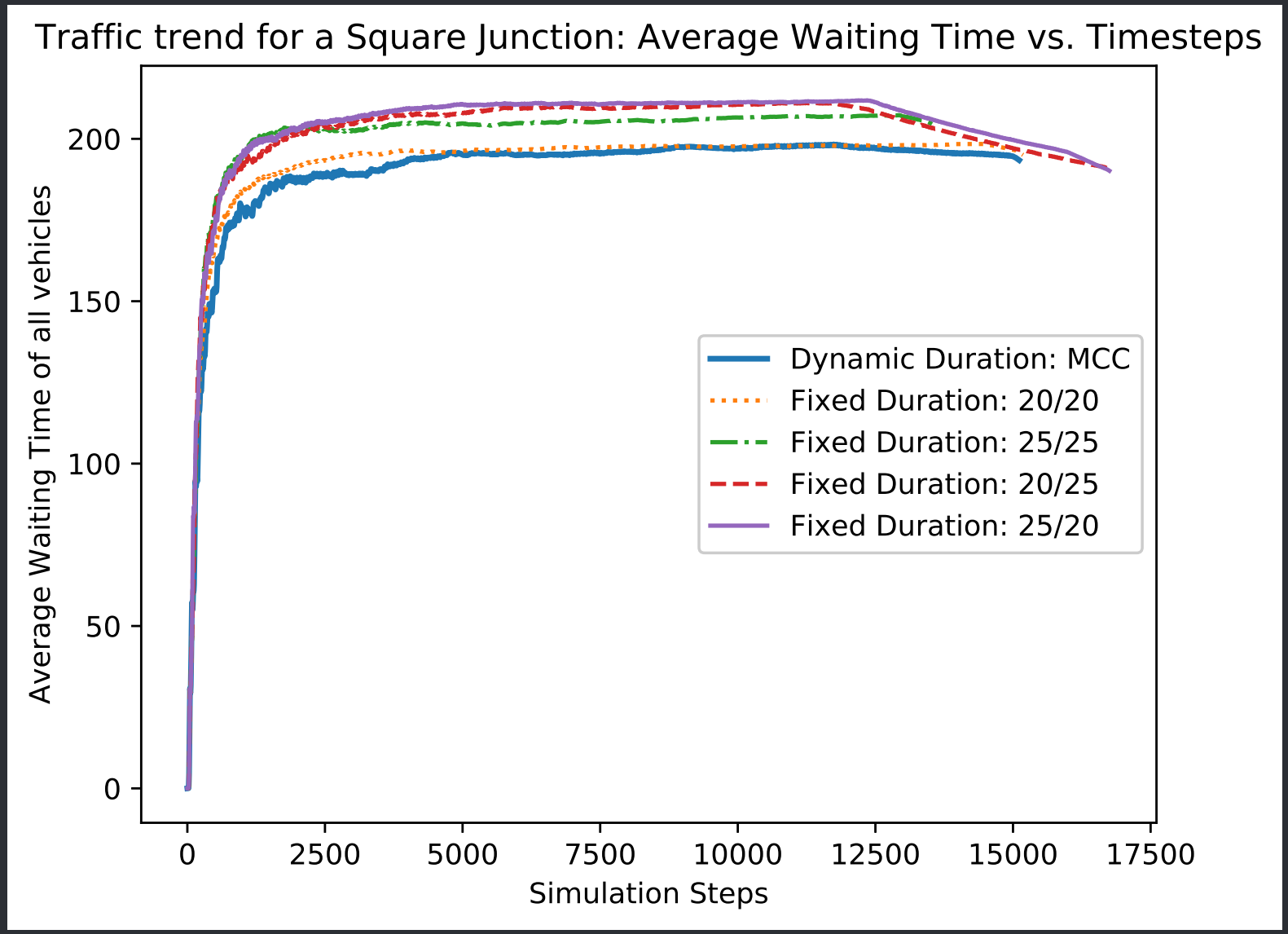
- Q-values: *Random numbers* $\in (0, 1]$
- Φ matrix, θ vector: *Random numbers* $\in (0, 1]$
- Number of vehicles: 22359
- Episode length (T): 300 timesteps
- Exploration factor (ϵ): 0.2
- Discount factor (γ): 0.9
- Step size (α): 0.075(QLFS) and 0.05(QLFA)

Section 5

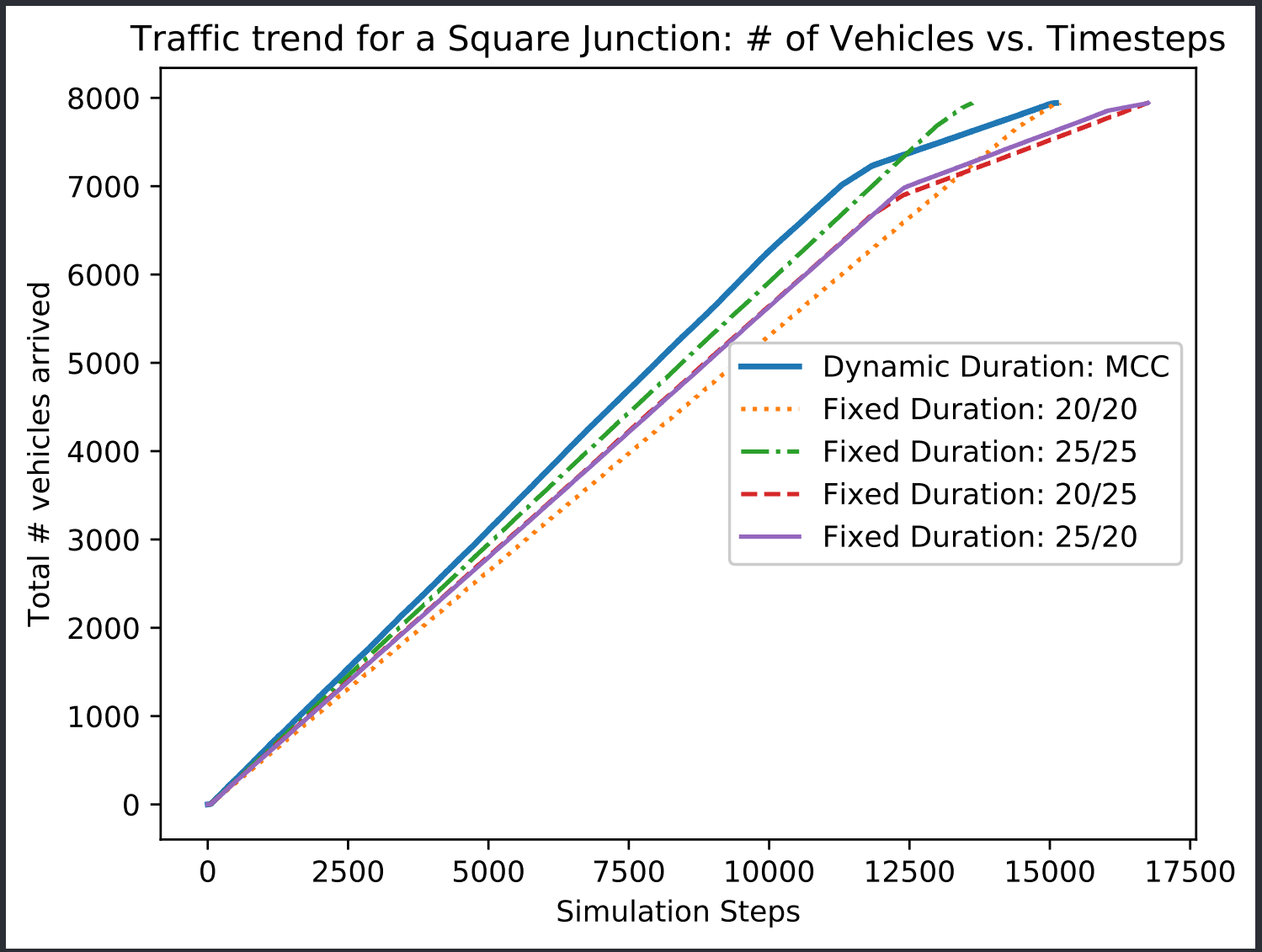
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Previous results

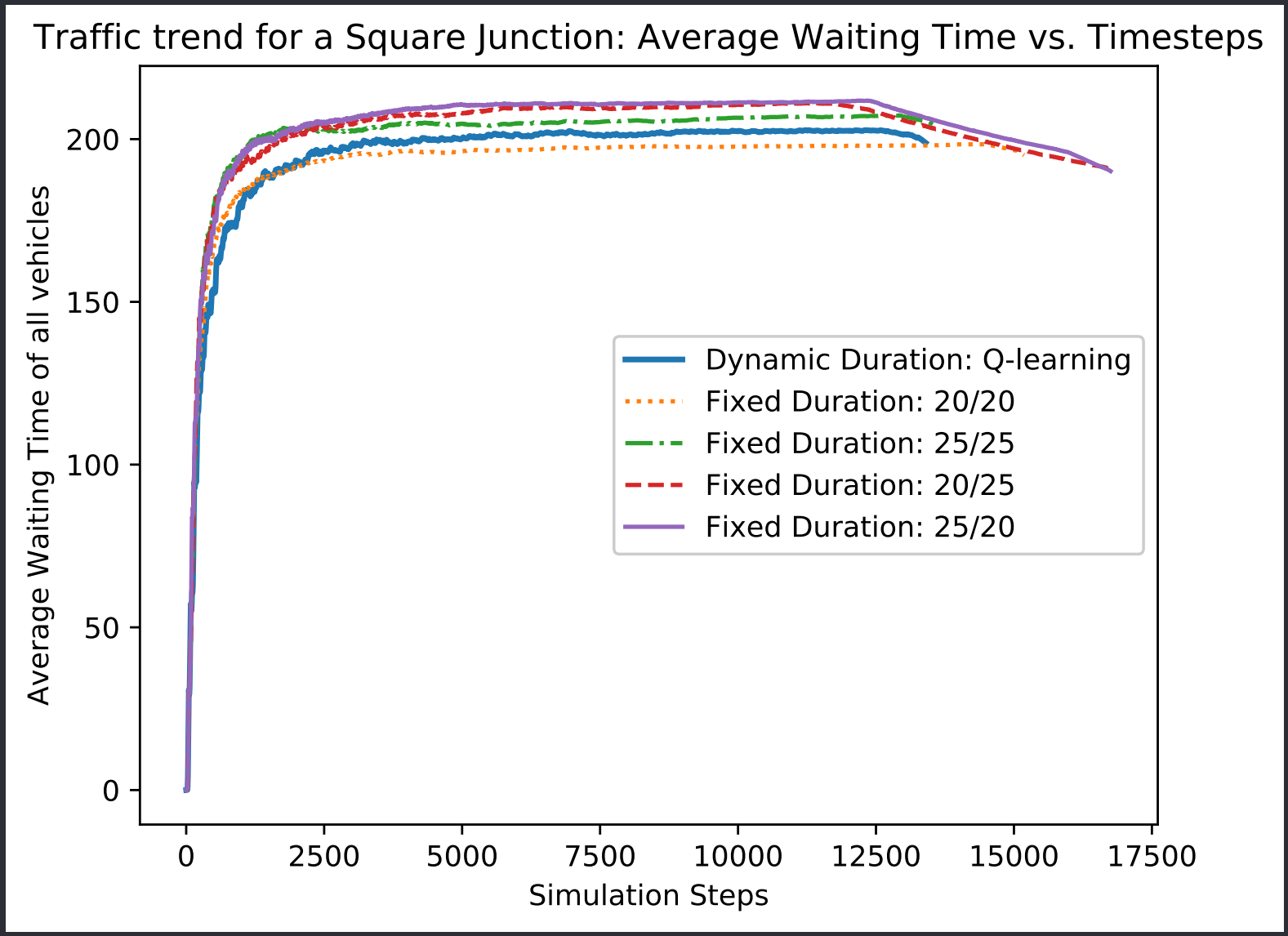
MCC: Average Waiting Time



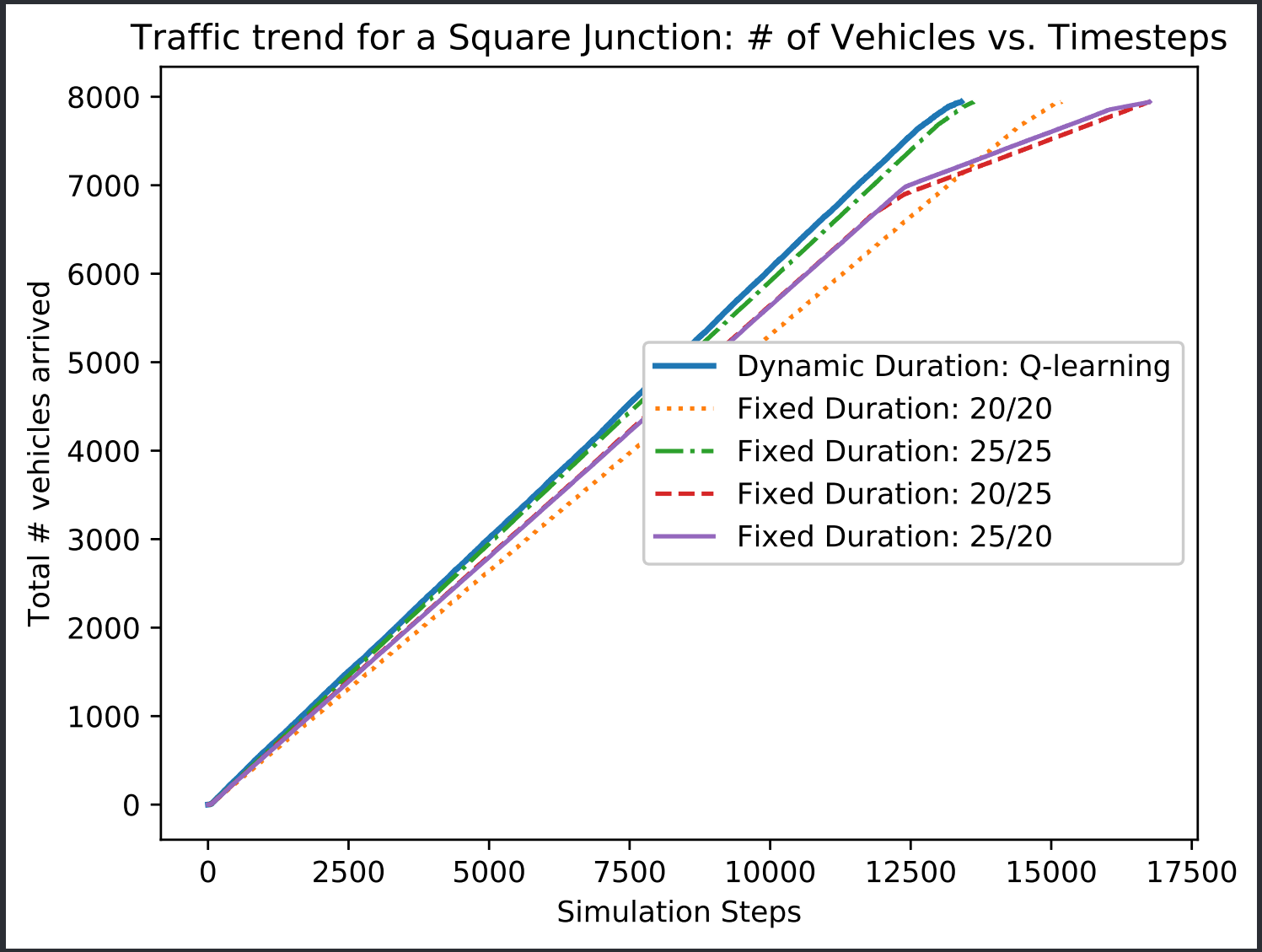
MCC: Total Number of Vehicles



Q-learning: Average Waiting Time



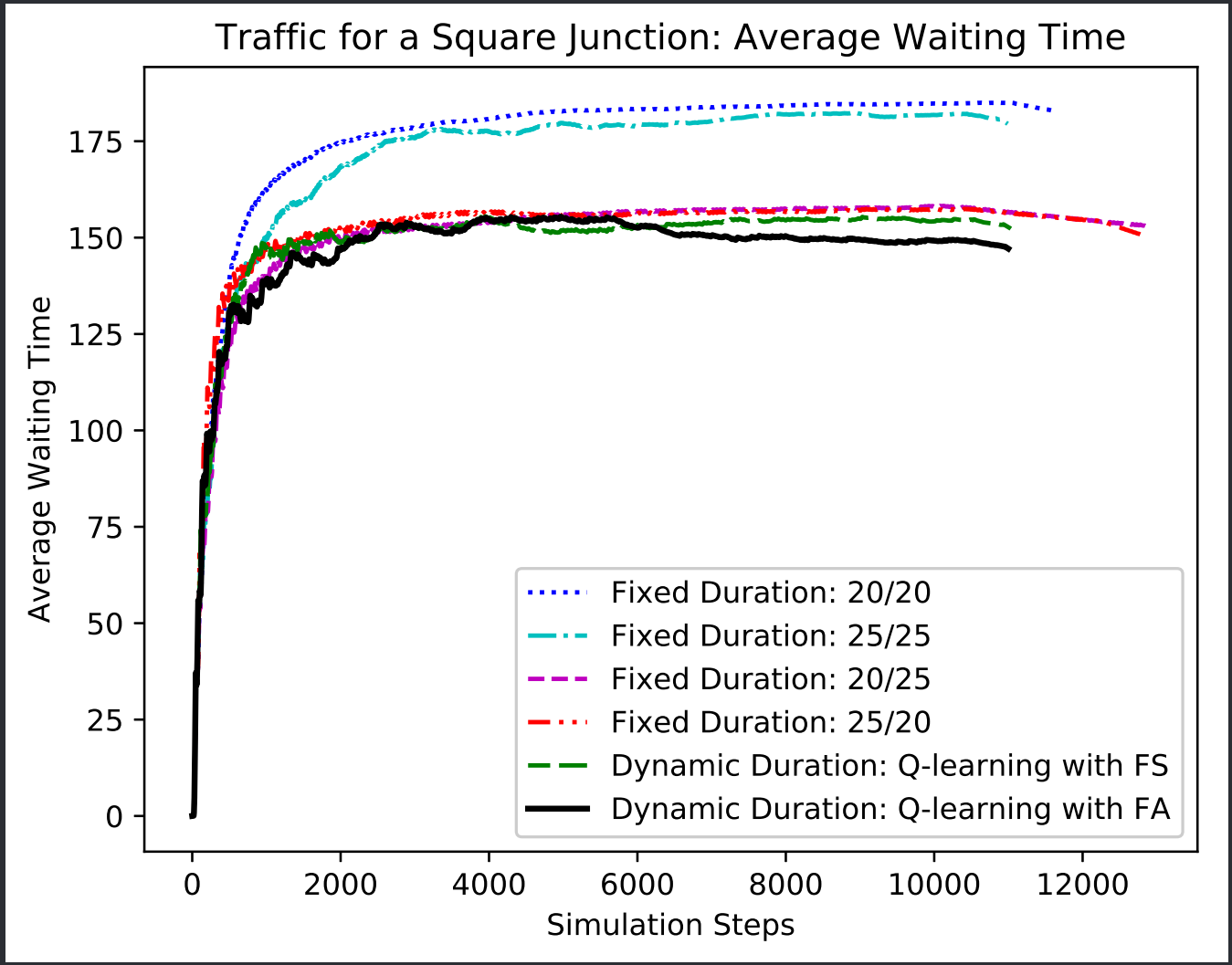
Q-learning: Total Number of Vehicles



Current results

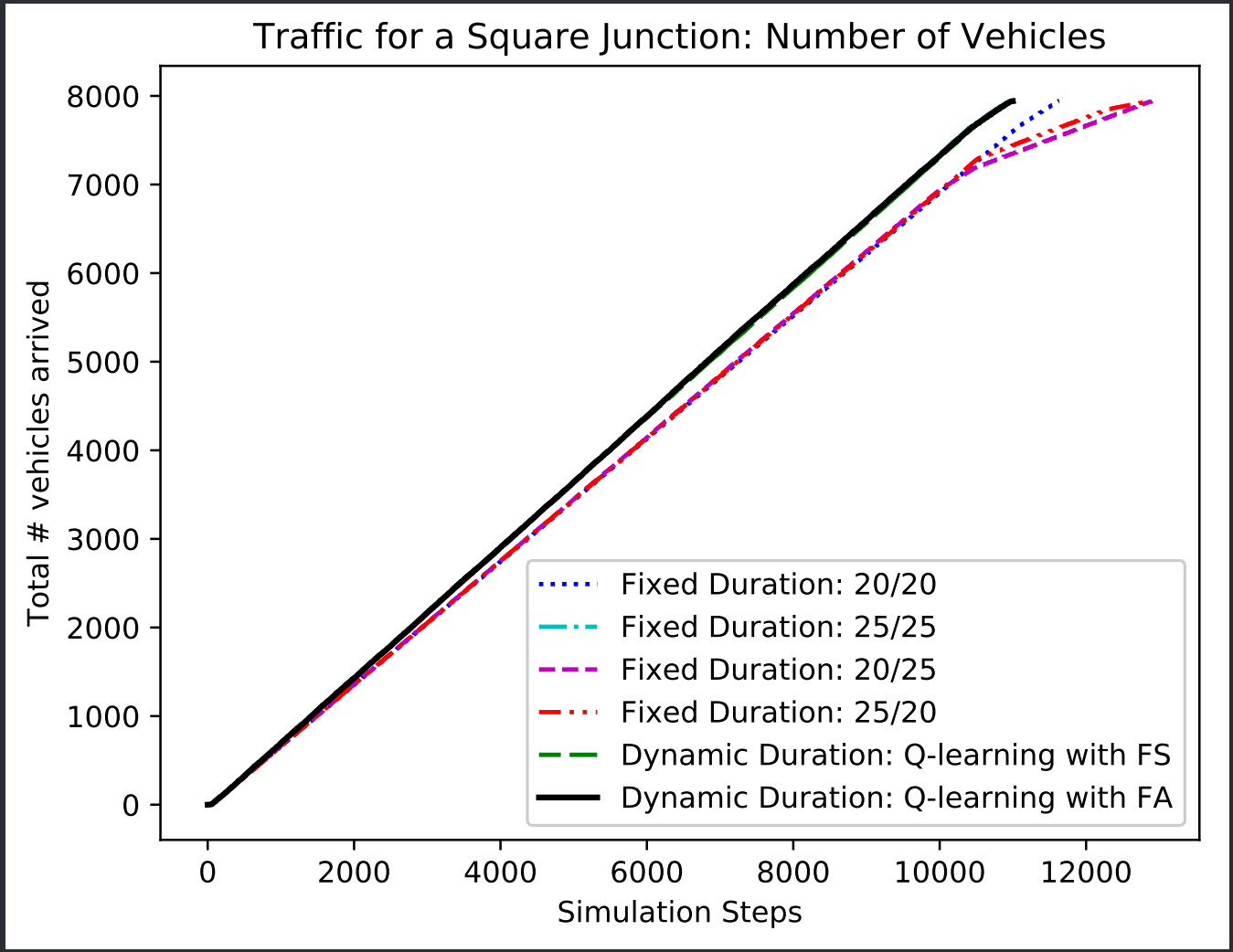
Scenario 1: a simple junction

Average Waiting Time



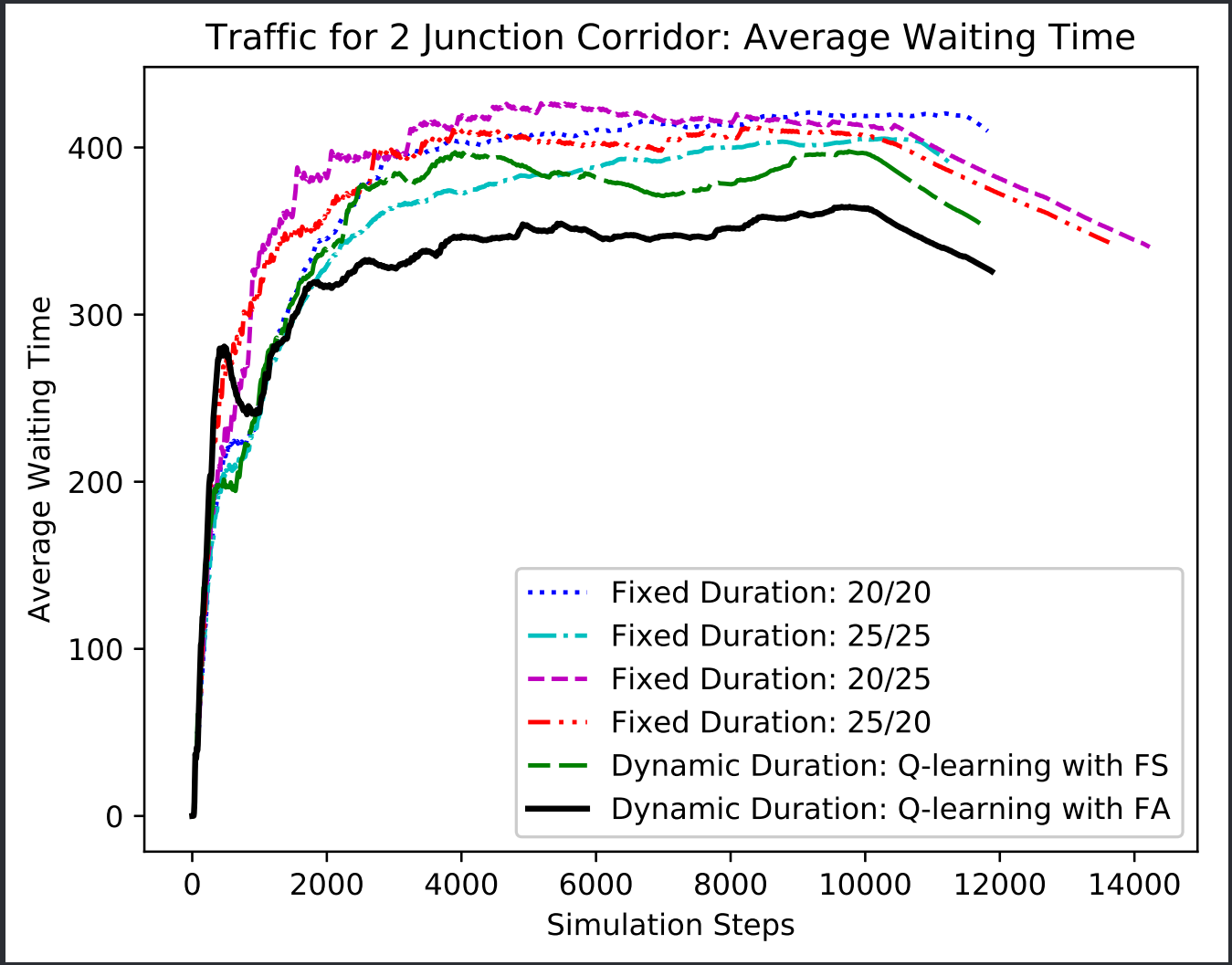
Scenario 1: a simple junction

Number of Vehicles



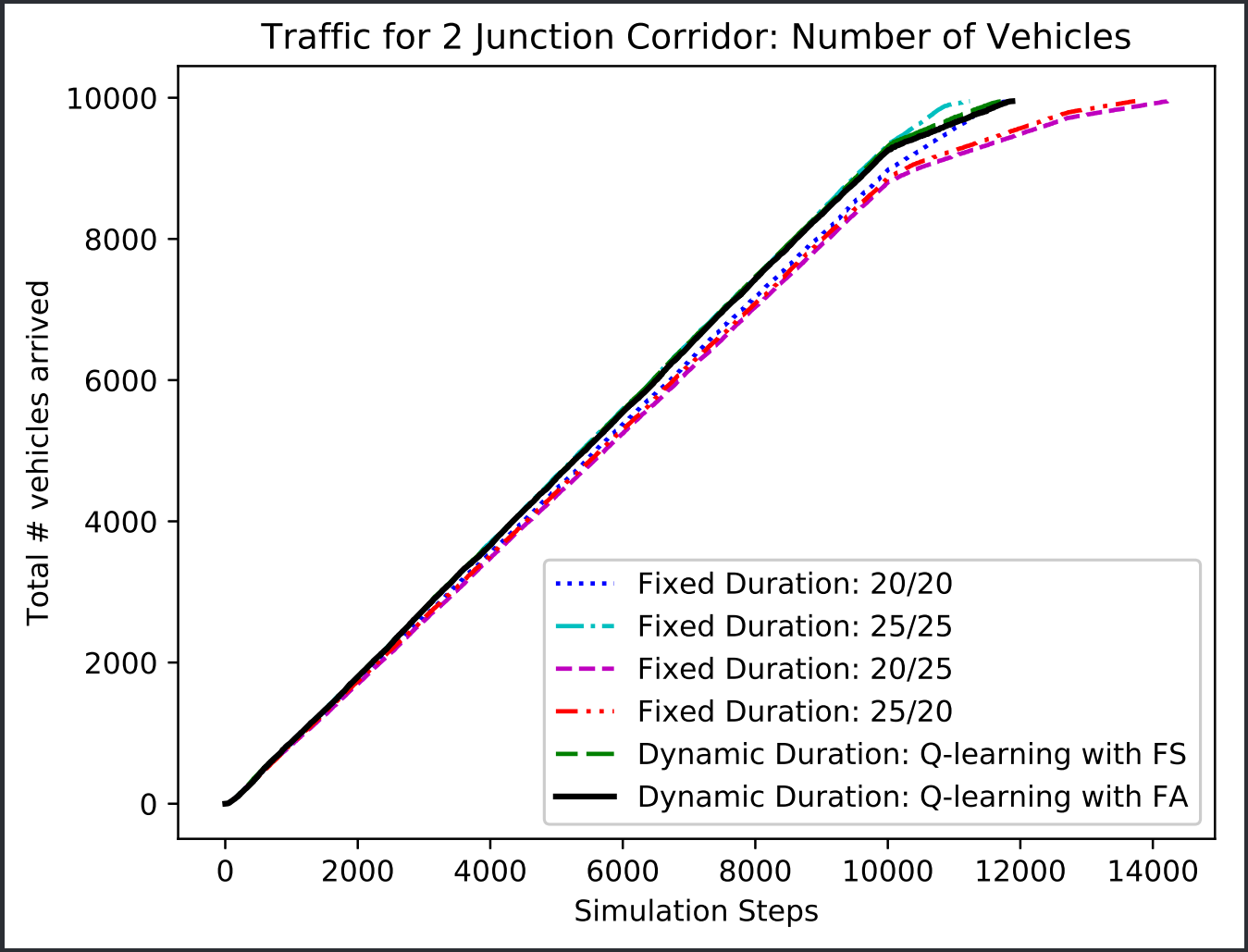
Scenario 2: two-junction corridor

Average Waiting Time



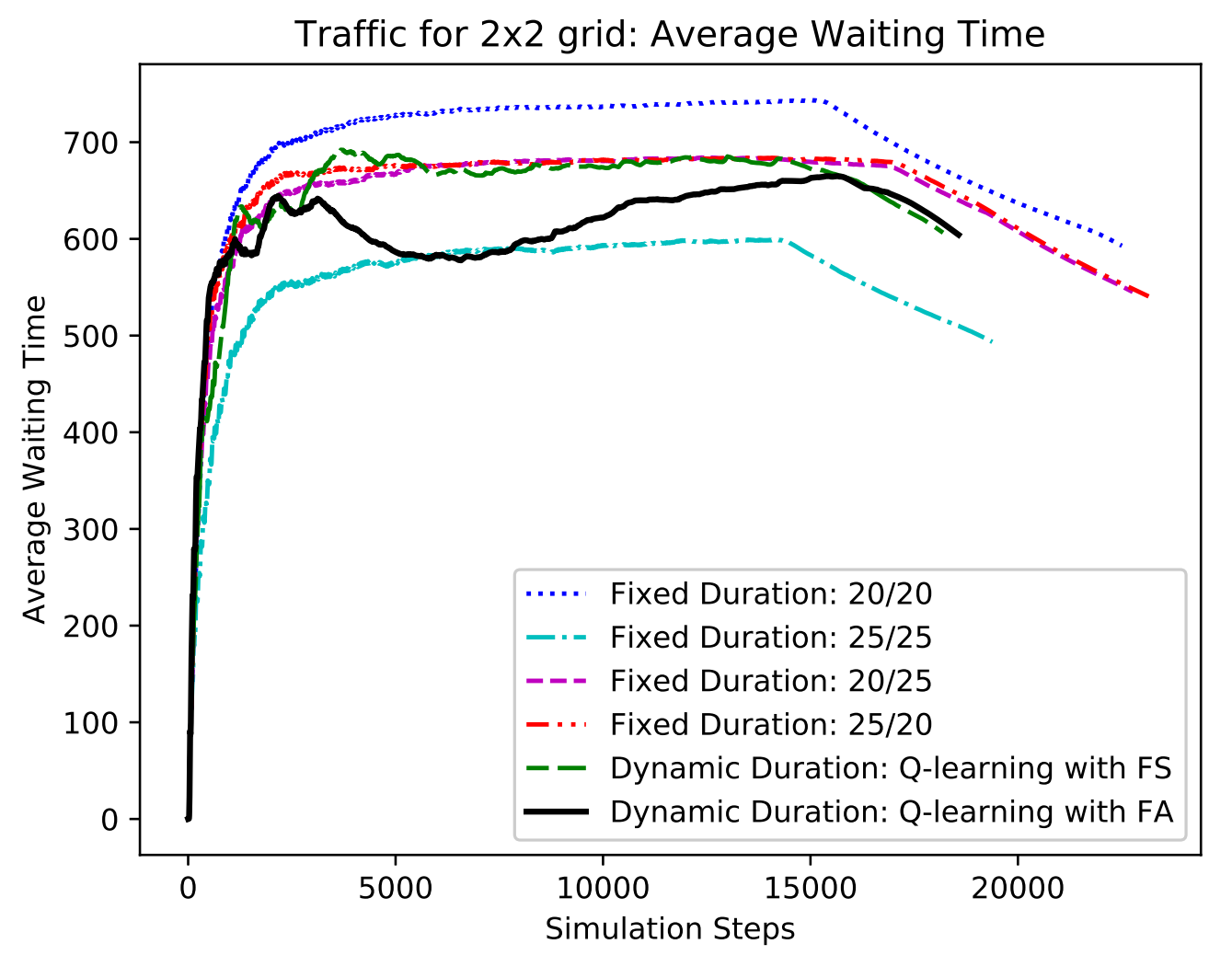
Scenario 2: two-junction corridor

Number of Vehicles



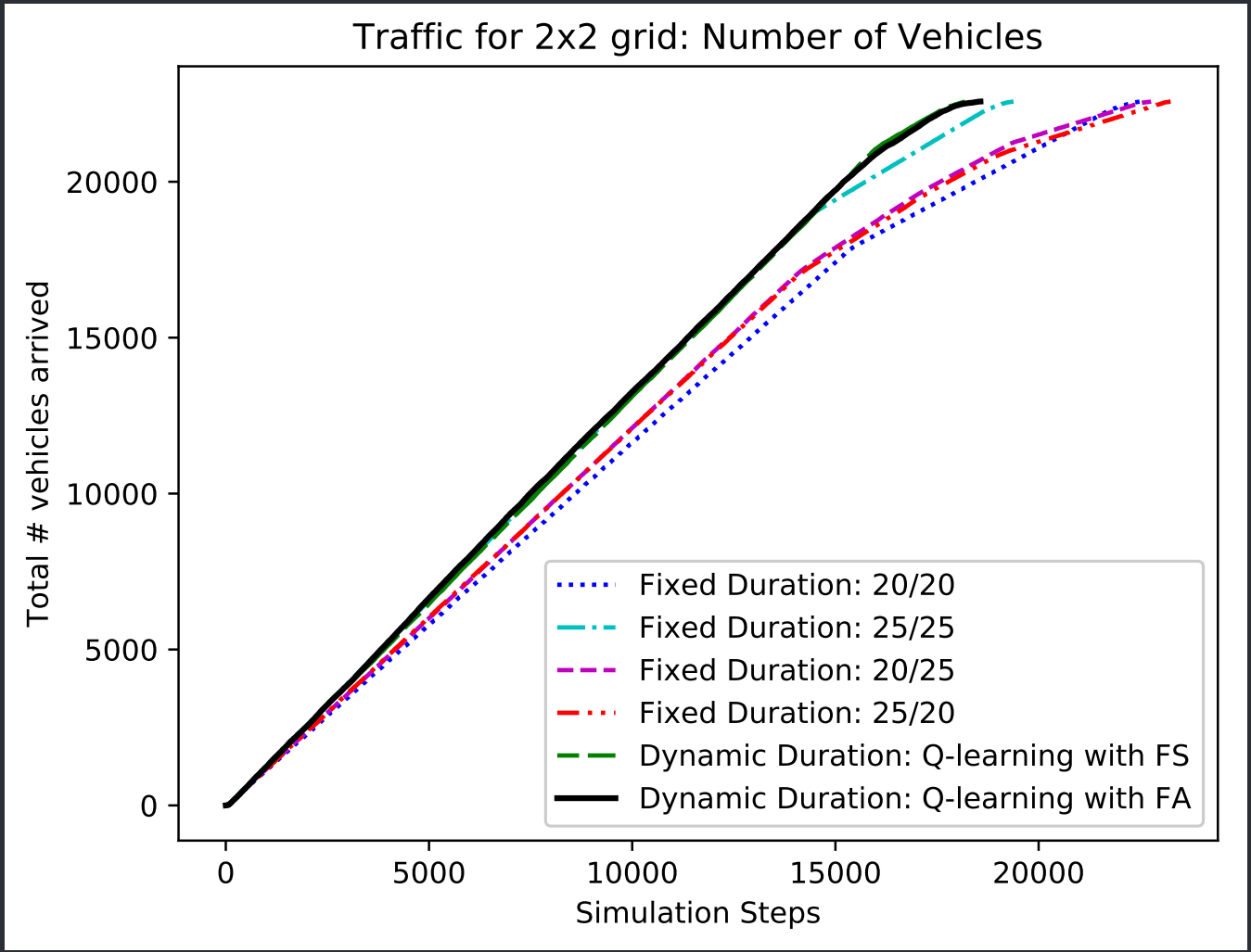
Scenario 3: a 2×2 grid

Average Waiting Time



Scenario 3: a 2×2 grid

Number of Vehicles



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Conclusions

- Traffic management is tricky but RL seems an interesting approach for TSC.

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Conclusions

- Traffic management is tricky but RL seems an interesting approach for TSC.
- We implemented two types of RL algorithms and evaluated them.
- Dynamic algorithms outperform their traditional counterparts.

Future Work

- ☑ More experiments.
- ☑ Extension of Q-learning based approach.
- ☑ Different types of road networks.
- ☐ Different types of vehicles.
- ☐ Investigation of policy gradient methods.

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Fin.

Thank you!